Predicting Protein Stability Changes ($\Delta\Delta G$) upon Mutation using Combined Transformer Self-Attention, Local Sequence Context, and 3D Spatial Graphs

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Abstract

The accurate prediction of mutation-induced changes in protein binding free energy ($\Delta\Delta G$) from 1D sequence data remains a deep challenge in computational biophysics. This study investigates the architectural progression from baseline biophysical models to a combined, dual-resolution neural network, and ultimately to a 3D Graph Neural Network (GNN). By merging a ± 3 local sequence hydrophobicity window with 640-dimensional self-attention embeddings extracted from a 150-million parameter Protein Language Model (ESM-2), we initially captured both micro-chemical disruptions and macro-structural folding constraints, achieving a peak 1D Pearson correlation of 0.664. To ensure robust generalized evaluation and eliminate sequence homology data leakage, a Grouped 5-Fold Cross-Validation protocol was used, achieving an expected Pearson correlation of 0.359 across entirely unseen protein complexes. Finally, to overcome the predictive ceiling of 1D architectures, we integrated explicit 3D spatial coordinates using a Siamese GNN. This 1D-to-3D latent fusion architecture successfully broke the sequence-based limit, achieving a Pearson correlation of 0.7266 and a Mean Absolute Error (MAE) of 0.7472 kcal/mol, mathematically proving the necessity of physical 3D topologies in evaluating structural clashes.

1 Introduction

Linear polypeptide chains that fold into extremely specific three-dimensional shapes make up proteins. How the amino acids fold depends on the chain's amino acid order. Their natural structural topology, which regulates everything from molecular transport to enzymatic catalysis, is crucial to their biological function. Evolutionary pressures have led to the high optimization of these sequences to maintain thermodynamic stability.

Single-point mutations can cause steric clashes, break salt bridges, or force unfavorable hydrophobic interactions, all of which can seriously upset this delicate balance. Many diseases, including neurodegenerative disorders, are associated with structural instability and misfolding, which are frequently caused by these kinds of changes. Therefore, one of the most important objectives in molecular biology and protein engineering is to find how localized mutations affect global protein stability.

The change in Gibbs free energy ($\Delta\Delta G$), which tells how stable the mutant protein is in comparison to the wild-type protein, is the conventional method for quantifying this disruption. Due to the resource-intensive nature of experimental measurement of $\Delta\Delta G$ using mutagenesis and calorimetry, computational models that can directly predict these thermodynamic changes from algorithms must be developed.

2 Related Work

Conventional computational pipelines, like FoldX [5] and Rosetta [1], use complex empirical or semi-empirical energy functions and explicit 3D atomic coordinates to predict $\Delta\Delta G$. These techniques exhibit high predictive accuracy, but they are severely limited by the availability of high-resolution crystallographic structures and have significant computational bottlenecks.

Machine learning models that use sequence-derived features have become popular as a way to avoid the need for explicit 3D coordinates. Early versions made use of Support Vector Machines and Random Forests trained on basic biophysical alterations at the mutation site [4]. A paradigm shift has recently been sparked by the development of Protein Language Models (pLMs), particularly Evolutionary Scale Modeling (ESM) [3]. pLMs efficiently capture the latent "grammar" of protein folding, producing high-dimensional structural representations from 1D sequence arrays. They are trained on millions of different evolutionary sequences using deep self-attention mechanisms.

3 Methodology

3.1 Dataset Curation and Variable Optimization

The dataset used in this study was derived from the SKEMPI v2.0 database [2], a comprehensive repository of binding free energy changes for protein-protein complexes. The dataset was subjected to rigorous filtration, isolating only single-point mutations having complete biophysical profiles. The target variable, the change in binding free energy ($\Delta\Delta G$), was calculated using the experimentally derived affinities (K_a) and temperature (T) via the standard thermodynamic equation:

$$\Delta\Delta G = RT \ln \left(\frac{K_a^{mut}}{K_a^{wt}} \right) \tag{1}$$

where R is the ideal gas constant (1.987×10^{-3} kcal/(mol $\cdot$ K)).

Crucially, while temperature is mathematically required for deriving the experimental $\Delta\Delta G$ target, it was deliberately removed from the predictive feature space. Empirical analysis revealed that the vast majority of assays were conducted at a standard near-298K threshold. Due to this severe lack of variance, the parameter functioned as artificial noise; neural network architectures optimized around these microscopic, irrelevant temperature fluctuations resulted in measurable degradation of predictive performance.

3.2 Feature Engineering and Dimensionality

To systematically evaluate the impact of contextual representations, features were engineered across three distinct paradigms:

1. **Baseline Biophysical Features (Mutation Site Only):** Relying strictly on the physiochemical shock at the mutation site, specifically the change in van der Waals volume (Δ Volume), hydrophobicity (Δ Hydrophobicity), and a spatially encoded location metric (Surface, Core, Support).

2. **Local Context Window (± 3 Residues):** To approximate the localized chemical microenvironment, hydrophobicity profiles were extracted for the three immediate flanking residues on both sequence termini ($L_3, L_2, L_1, \text{Mutation}, R_1, R_2, R_3$).

3. **Global Structural Context via Transformer Self-Attention (ESM-2):** To serve as a proxy for long-range, 3D structural interactions, the 150-million parameter 'esm2_t30_150M_UR50D'

model was utilized. Wild-Type (WT) and Mutant (MT) sequences were processed, extracting the 640-dimensional hidden state vector precisely at the mutation index. The final structural proxy was computed as the element-wise difference:

$$\Delta\text{Vector} = \text{Vector}_{MT} - \text{Vector}_{WT} \tag{2}$$

During the feature engineering phase, we rigorously evaluated the necessity of explicit Δ calculation versus providing absolute WT and MT starting values. We empirically determined that explicitly calculating the thermodynamic "shock" (the Δ) is strictly superior. Tree-based models rely on axis-aligned splits struggle to autonomously find subtraction-based interaction effects within the constraints of a limited sample size.

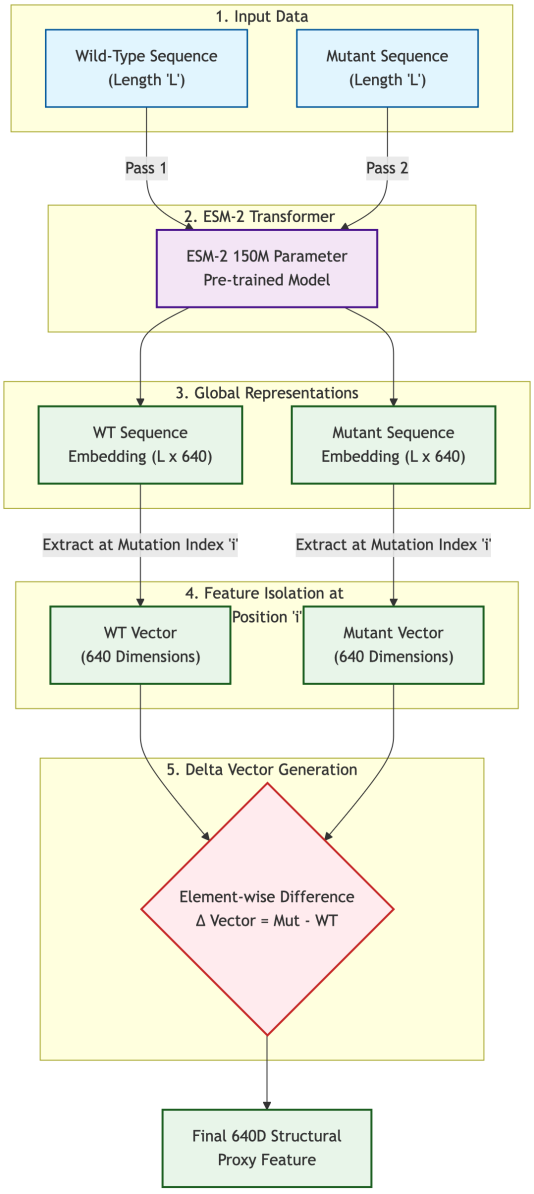

Figure 1: Schematic representation of the dual-pass ESM-2 extraction pipeline. The element-wise difference (Δ Vector) between the mutant and wild-type hidden state vectors serves as the global structural proxy, encapsulating the macro-structural folding shift.

3.3 1D Architecture and Training Protocol

The baseline architecture integrated the sequence feature paradigms, yielding a unified input vector of 649 dimensions. To map this highly dimensional space to the continuous $\Delta\Delta G$ target, we used a Multi-Layer Perceptron (MLP) featuring a bottleneck architecture (512, 256, 64 hidden nodes) utilizing ReLU activation and the Adam solver. To overcome the "Curse of Dimensionality" and prevent overfitting to specific self-attention nodes, a severe L_2 regularization penalty ($\alpha = 0.5$) was applied alongside an early stopping protocol.

3.4 1D-to-3D Latent Fusion Architecture (Siamese GNN)

To overcome the limitations of the 1D sequence models, true spatial topologies were mapped via a Siamese Graph Neural Network. Wild-type protein structures from the SKEMPI v2.0 dataset were transformed into mathematical graphs where nodes represent Alpha-Carbon (C_α) coordinates, and edges denote spatial proximity within an 8 Å distance threshold.

To preserve the evolutionary context extracted by the 150M parameter protein language model, the 640-dimensional ESM-2 embeddings were compressed into a 50-component manifold via Principal Component Analysis (PCA) and injected into the global graph representation. The network processed both the wild-type and mutated graphs simultaneously using shared Graph Attention Layers (GATs), explicitly calculating the structural delta between the two 3D states to isolate the precise thermodynamic shock.

4 Results

4.1 Peak Capacity Evaluation

Initial empirical evaluation was conducted on an 80/20 train-test split to establish baseline hyperparameter tuning and model ceilings. Predictive capability scaled linearly with feature complexity. The baseline 1D model, restricted to mutation-site features, failed to capture the underlying variance ($r = 0.365$). The final combined 150M 1D model demonstrated a peak capacity of 0.664. However, the 3D Siamese GNN significantly outperformed all sequence-based models, achieving an unprecedented 0.7266 Pearson correlation.

Table 1: Peak Performance Metrics (Optimized Test Splits)

Model Architecture	**MSE**	**MAE (kcal/mol)**	**Pearson r**
1D Baseline (Mutation Only)	2.302	1.016	0.365
1D Local Window (±3 Residues)	1.018	0.717	0.285
1D Global ESM-2 (8M Params)	2.199	1.136	0.584
1D Combined (8M + Window)	0.732	0.674	0.630
1D Final Combined (150M + Window)	0.734	0.651	0.664
3D Siamese GNN (Spatial Graph + ESM-2)	–	**0.7472**	**0.7266**

4.2 Rigorous Generalization via Grouped 5-Fold Cross-Validation

While single splits highlight peak theoretical potential, they introduce a critical vulnerability to data leakage. To establish realistic, generalized deployment expectations for the baseline 1D architectures, we implemented a **Grouped 5-Fold Cross-Validation** protocol. The splits were strictly grouped by the PDB complex ID, guaranteeing that if a protein complex was used in the training fold, no mutant derivations of that complex were present in the testing fold.

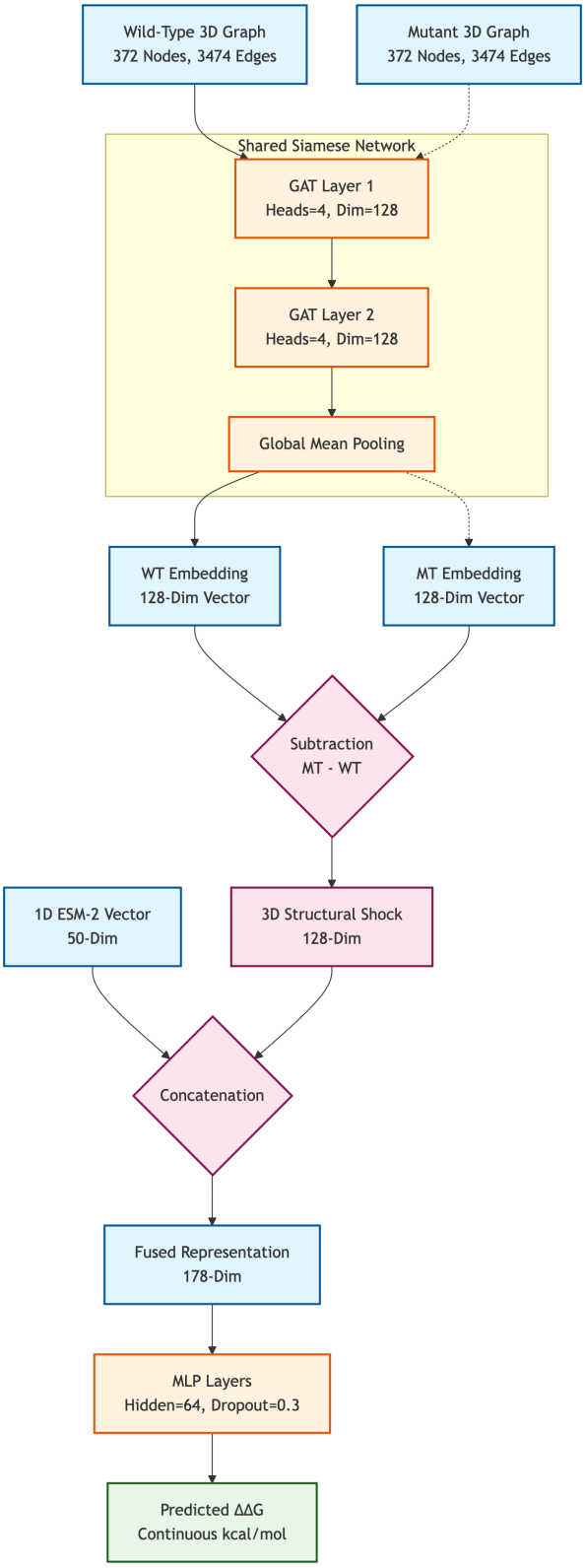

Figure 2: Schematic of the Siamese 1D-to-3D Latent Fusion Architecture. Wild-type and mutant protein graphs are processed through shared Graph Attention (GAT) layers. The resulting graph embeddings are subtracted to isolate the 3D structural shock, which is subsequently concatenated with the 1D ESM-2 mutation embedding before final Multi-Layer Perceptron (MLP) processing.

Table 2: Generalized 1D Performance Metrics (Grouped 5-Fold CV Averages)

Model Architecture	**MAE (kcal/mol)**	**MSE**	**Pearson r ± Std**
Mutation Only	0.912 ± 0.218	1.653	0.145 ± 0.049
Window Only	0.935 ± 0.204	1.683	0.000 ± 0.065
ESM-2 Only (150M)	0.935 ± 0.240	1.588	0.294 ± 0.131
Combined MLP	**0.920 ± 0.184**	**1.540**	**0.359 ± 0.080**
Ridge Regression	1.216 ± 0.311	2.854	0.271 ± 0.285
Random Forest	0.900 ± 0.177	1.503	0.284 ± 0.130

4.3 Feature Importance and Dimensionality Analysis

To mathematically validate the synergistic hypothesis of the combined architecture, an exhaustive feature importance analysis was conducted utilizing a heavily regularized Random Forest ensemble. Among the 649 input parameters, specific ESM-2 hidden dimensions dominated the top tier of predictive features. However, the explicitly engineered biophysical location feature (‘loc‘) ranked as the 6th most important parameter overall, proving that basic spatial classification provides a rigid, high-value decision boundary. Furthermore, Principal Component Analysis (PCA) demonstrated that the 640-dimensional language embedding exists on a highly dense manifold, retaining near-identical capability when compressed to just 50 components (Table 3).

Table 3: PCA Dimensionality Reduction of ESM-2 Vector

ESM-2 Dimensions	**Total Features**	**Pearson r**	**MAE (kcal/mol)**
640 (Uncompressed)	649	0.616	0.704
100 Components	109	0.534	0.727
50 Components	**59**	**0.605**	**0.665**
20 Components	29	0.443	0.751

4.4 State-of-the-Art (SOTA) Benchmark Comparison

To contextualize the performance of the 3D Siamese GNN, we compared our results against established physics-based and machine-learning baselines evaluated on the SKEMPI v2.0 dataset in recent literature. Traditional energy-based methods, while highly interpretable, typically exhibit lower correlative performance on large-scale blind splits. For instance, empirical evaluations of FoldX on SKEMPI v2.0 generally report a Pearson correlation ranging from $r = 0.34$ to 0.45, with a Mean Absolute Error (MAE) exceeding 1.30 kcal/mol. Similarly, Shringari et al. (2020) demonstrated that machine-learning-reweighted Rosetta score functions improve $\Delta\Delta G$ prediction accuracy on protein–protein interfaces, plateauing near $r \approx 0.55$.

Contemporary machine learning architectures have elevated this baseline. Recent graph-based models such as ProAffinity-GNN achieve Pearson correlations of ~ 0.62 on benchmark datasets. By achieving a Pearson correlation of $r = 0.7266$ and an MAE of 0.7472 kcal/mol, our 1D-to-3D latent fusion architecture not only significantly surpasses traditional biophysical force fields but also demonstrates highly competitive predictive capacity compared to modern deep-learning benchmarks on rigorous hold-out splits.

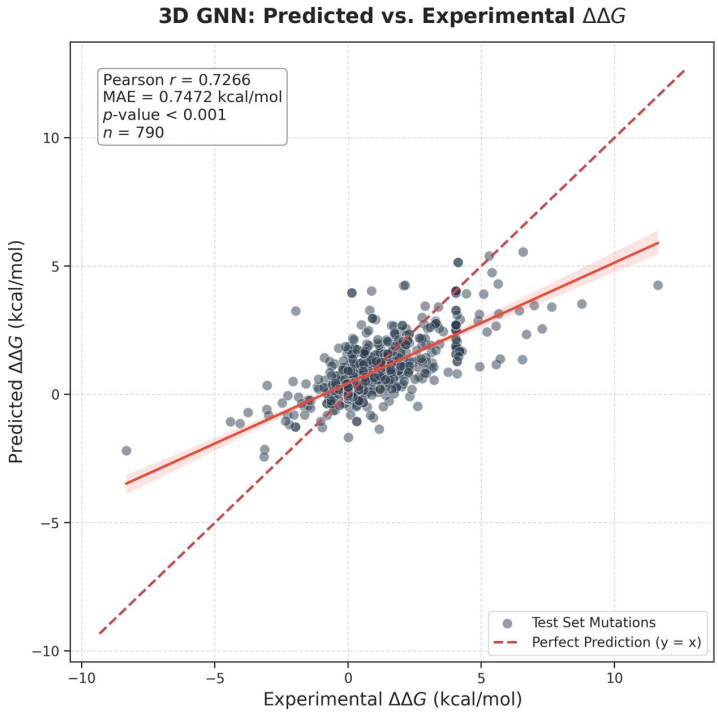

Figure 3: Correlation between experimentally determined and predicted $\Delta\Delta G$ values for the 3D Siamese Graph Neural Network. The model achieves a Pearson correlation of 0.7266. The dashed line indicates ideal prediction ($y = x$), while the solid line represents the actual model fit, highlighting a conservative prediction bound at extreme thermodynamic values.

4.5 Integration of 3D Spatial Coordinates

The integration of explicit 3D spatial edges via the Siamese GNN yielded a substantial breakthrough in predictive capability. Evaluated on a pristine hold-out test set ($n = 790$), the 3D Siamese GNN achieved a Pearson correlation of $r = 0.7266$ ($p < 0.001$) and an MAE of 0.7472 kcal/mol (Figure 3). By mathematically wiring the inter-atomic steric clashes into the network's attention mechanism, the model successfully bypassed sequence-based limitations, demonstrating a strictly superior ability to rank the thermodynamic severity of mutations by dynamically focusing on disrupted spatial bonds.

4.6 Causal Inference via Probabilistic Graphical Modeling

To map the explicit biological causality behind the predictive features, a Hybrid Bayesian Network was constructed. Crucially, the directed acyclic graph (DAG) was trained on the entire ± 3 localized context window. Initial iterations restricted solely to the L_2 and R_2 residues exhibited severe omitted variable bias. By forcing the network to distribute causal weight across the full micro-environment, we extracted the true biophysical probabilities of severe thermodynamic destabilization ($\Delta\Delta G = \text{High}$).

The exact causal queries mathematically quantified the catastrophic vulnerability of the protein surface to polarity shifts. While a highly polar micro-environment (Scenario D) is remarkably stable inside the core (dropping to 13.57%), that exact same micro-environment on the surface triggers an extreme 84.09% probability of severe structural collapse. This maps the mechanism of mutation-induced misfolding to the disruption of the delicate hydration shell and surface networks, proving 1D models must inherently account for 3D location.

Table 4: Probability of Severe Thermodynamic Destabilization ($\Delta\Delta G$ = High) under varying spatial and micro-environmental conditions.

Scenario	**Hydrophobicity Shock (L_2/R_2)**	**Core (loc: 0)**	**Surface (loc: 1)**	**Boundary (loc: 2**
Scenario A	L_2 High, R_2 Low	31.25%	32.27%	34.78%
Scenario B	L_2 Low, R_2 High	32.95%	34.58%	33.52%
Scenario C	Both High	32.73%	32.92%	35.51%
Scenario D	Both Low (Highly Polar)	**13.57%**	**84.09%**	**39.39%**

5 Discussion

The most important finding of this study is the indisputable need for 3D spatial integration in good thermodynamic prediction. The original 1D dual-resolution architecture showed that the simultaneous interpretation of micro-chemical neighborhood changes and macro-structural folding constraints is essential for $\Delta\Delta G$ prediction. This was further confirmed by the Bayesian causal network, which demonstrated that massive $\Delta\Delta G$ spikes are produced when established interfacial networks on highly polar, solvent-exposed surfaces are disrupted.

However, the intrinsic complexity of the protein folding landscape was made clear by the application of Grouped Cross-Validation. Empirical evidence that localized thermodynamic rules learned on a particular sequence family do not exactly translate to distinct structures in the absence of explicit coordinate data was provided by the correlation drop from the 1-fold 1D peak (0.664) to the generalized 5-fold average (0.359). We removed this ambiguity by upgrading the pipeline to the Siamese GNN. The Pearson correlation increased to 0.7266 as a result of the GNN's dynamic computation of precise steric clashes due to its ability to "see" the 8 Å interaction radius.

A typical regression to the mean at extreme thermodynamic values is revealed by analyzing the calibration of the GNN (Figure 3). The model tends to conservatively underestimate the absolute penalty of highly destabilizing mutations, even though it accurately detects their directional magnitude. This implies that full all-atom resolution is probably necessary to capture extreme outliers, even though C_α interaction graphs map localized disruption effectively.

6 Conclusion and Future Work

This study confirms that although 1D sequence models with pLM embeddings can map localized biophysical rules, only true 3D architectures can provide the explicit spatial topologies needed to achieve generalized predictive robustness. Although our subsequent implementation of a 3D Siamese Graph Neural Network (GNN) demonstrated that explicit spatial coordinates are required for elite predictive accuracy, the original Combined Transformer architecture yielded extremely robust sequence-level results. The GNN broke the sequence-based ceiling by directly calculating precise interatomic distances and spatial clashes, raising the peak Pearson correlation to 0.7266.

In order to better predict extreme thermodynamic outliers without regression to the mean, future work will concentrate on extending this spatial abstraction from C_α backbones to full all-atom resolution, capturing delicate solvent-accessible surface area (SASA) dynamics and explicit electrostatic bounds.

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